Nets of Solids

Name _____ Date _____ Calculator allowed Version 1

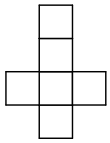
Scan to practise this topic online

KEY WORDS net *(raupapa mata)* · face *(mata)* · edge *(tapa)* · fold · cube *(mataono)* · prism *(poro)*

A Nets of a cube ●○○

Remember: A net is the flat shape that folds up into a solid. A cube needs six squares, but not every arrangement of six squares will fold: picture the squares closing round a cube.

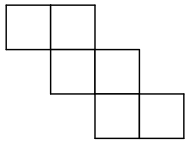
1. Does each arrangement fold into a cube? Write yes or no.

a) 

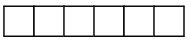
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b) 

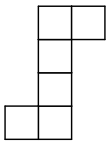
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c) 

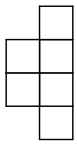
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d) 

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e) 

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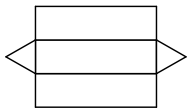
f) 

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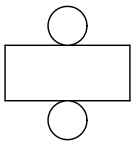
B Which solid? ●○○

Remember: Count the faces and look at their shapes: two triangles and three rectangles make a triangular prism; a rectangle and two circles make a cylinder.

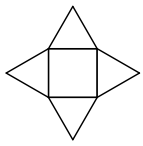
2. Name the solid that each net folds into.

a) 

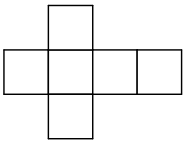
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b) 

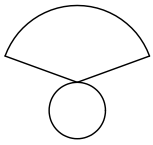
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c) 

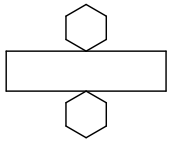
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d) 

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e) 

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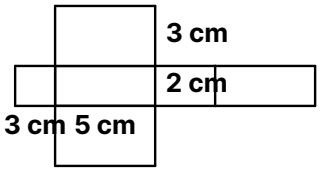
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C Nets of cuboids ●●○

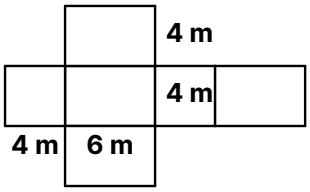
> **Remember:** Each face appears once in the net. Opposite faces are the same size, so a cuboid net has three pairs of rectangles. Surface area = the area of the whole net.

3. Each net folds into a cuboid. Find the total area of the net (the surface area of the cuboid).

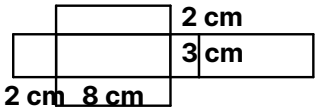
a)



b)



c)

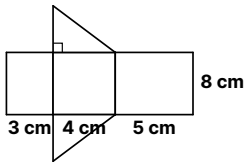


D Nets of prisms ●●●

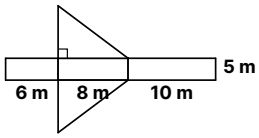
> **Remember:** A triangular prism has two triangles and three rectangles. Each rectangle is as wide as one side of the triangle and as long as the prism.

4. Each net folds into a triangular prism. Find the total area of the net.

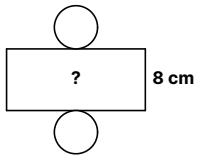
a)



b)



5. The net of a cylinder of radius 3 cm and height 8 cm is a rectangle and two circles.



a) How long is the rectangle? Give the answer to 1 decimal place.

b) Find the total area of the net, to 1 decimal place.

E Drawing nets ●●●

> **Remember:** Draw the faces full size with a ruler, joined along edges so the shape folds without gaps or overlaps. Label every length.

6. Draw an accurate net for each solid on paper, using the lengths given (in centimetres). Then calculate the area of your net.

a) A cuboid 4 cm by 3 cm by 2 cm.

b) A triangular prism whose end is a right-angled triangle with sides 3 cm, 4 cm and 5 cm, and whose length is 2 cm.

★ Extension

7. An open box (no lid) is made from a square sheet of card 10 cm by 10 cm by cutting a 2 cm square from each corner and folding up the sides. Sketch the net, and find the volume of the box.

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8. How many different nets does a cube have? Sketch two that are not just rotations or reflections of each other, and explain why a 2 by 3 block of six squares is not one of them.

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Nets of Solids

Name _____ Date _____ Calculator allowed Version 2

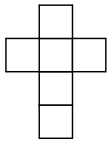
Scan to practise this topic online

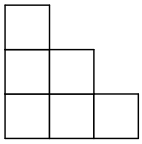
KEY WORDS net *(raupapa mata)* · face *(mata)* · edge *(tapa)* · fold · cube *(mataono)* · prism *(poro)*

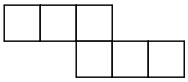
A Nets of a cube ●○○

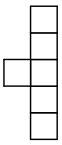
Remember: A net is the flat shape that folds up into a solid. A cube needs six squares, but not every arrangement of six squares will fold: picture the squares closing round a cube.

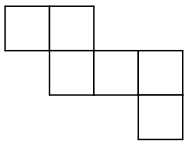
1. Does each arrangement fold into a cube? Write yes or no.

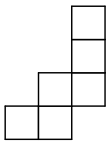
a)  _____

b)  _____

c)  _____

d)  _____

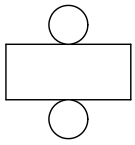
e)  _____

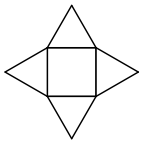
f)  _____

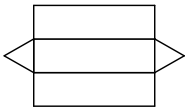
B Which solid? ●○○

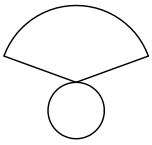
Remember: Count the faces and look at their shapes: two triangles and three rectangles make a triangular prism; a rectangle and two circles make a cylinder.

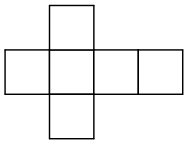
2. Name the solid that each net folds into.

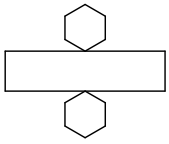
a)  _____

b)  _____

c)  _____

d)  _____

e)  _____

f)  _____

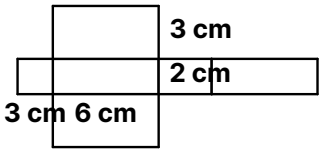
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C Nets of cuboids ●●○

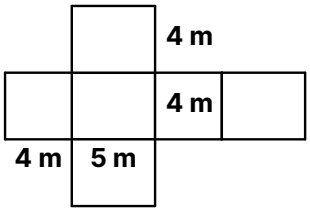
Remember: Each face appears once in the net. Opposite faces are the same size, so a cuboid net has three pairs of rectangles. Surface area = the area of the whole net.

3. Each net folds into a cuboid. Find the total area of the net (the surface area of the cuboid).

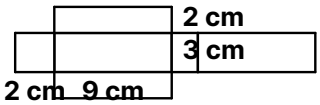
a)



b)



c)

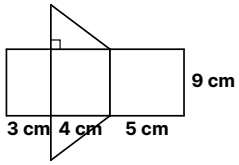


D Nets of prisms ●●●

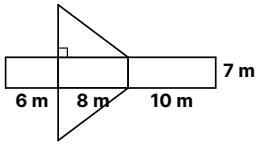
Remember: A triangular prism has two triangles and three rectangles. Each rectangle is as wide as one side of the triangle and as long as the prism.

4. Each net folds into a triangular prism. Find the total area of the net.

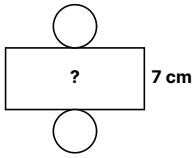
a)



b)



5. The net of a cylinder of radius 4 cm and height 7 cm is a rectangle and two circles.



a) How long is the rectangle? Give the answer to 1 decimal place.

b) Find the total area of the net, to 1 decimal place.

E Drawing nets ●●●

Remember: Draw the faces full size with a ruler, joined along edges so the shape folds without gaps or overlaps. Label every length.

6. Draw an accurate net for each solid on paper, using the lengths given (in centimetres). Then calculate the area of your net.

a) A cuboid 5 cm by 3 cm by 2 cm.

b) A triangular prism whose end is a right-angled triangle with sides 3 cm, 4 cm and 5 cm, and whose length is 3 cm.

★ Extension

7. An open box (no lid) is made from a square sheet of card 12 cm by 12 cm by cutting a 2 cm square from each corner and folding up the sides. Sketch the net, and find the volume of the box.

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8. How many different nets does a cube have? Sketch two that are not just rotations or reflections of each other, and explain why a 2 by 3 block of six squares is not one of them.

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